



Weekly Report

- **Title:** Fortnightly Training Report
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- **Program:** Bachelor of Computer Science
- **Week:** week 17 and week 18
- **Dates:** 7 Jun to 18 Jun 2026
- **Organization/Company Name:** Wadi Makkah Technology Company
- **Supervisor Name:** Mr. Khalid Abdulghani
- **Submission Date:** 18 Jun 2026

1. Introduction

During this training period, I continued working on the development and enhancement of the Soft Landing Platform, focusing on completing the system's operational functionalities, improving the user experience, and resolving technical issues related to localization, translation, and data integration. In addition, I explored automation and artificial intelligence technologies to develop solutions that support more efficient task execution and reduce manual intervention in certain processes.

2. Tasks and Responsibilities

Week 17:

- **n8n Workflow Automation Project:** Designed and developed an automated workflow using n8n, integrating a Telegram Bot with an AI Agent to receive, process, and respond to messages automatically, reducing manual intervention and improving communication efficiency.
- **AI and System Integration:** Built an end-to-end workflow that receives messages from Telegram, forwards them to an AI model for processing, and automatically delivers generated responses back to users, ensuring seamless data flow between integrated services.

- **Text Processing Automation:** Developed an automated text-formatting feature that converts the first letter of each word into uppercase (Title Case), improving text consistency and reducing the need for manual formatting.

- **Intersect OS Project – Soft Landing Platform**

The One-Time Password (OTP) verification feature was developed to enhance platform security during both registration and login processes. A verification code is sent to the user's email address and must be validated before account creation is completed or access to the system is granted. The implementation included a dedicated OTP verification page, an expiration timer with a resend option, and full integration between the user interface, backend services, and database to ensure a secure and seamless user experience.

Week 18:

Intersect OS Project – Soft Landing Platform

- Developed and enhanced the notification system to display important events for both users and administrators.
- Integrated notifications with key system operations such as new company registrations, document uploads, and status updates.
- Improved document management by implementing mechanisms for document review, approval, and resubmission requests.
- Developed the licensing pages and integrated them with the database to display finalized company licenses.
- Resolved localization and translation issues and added missing content in both the Arabic and English versions of the platform.
- Enhanced the automatic update process for request and stage statuses based on the status of their associated tasks.
- Reviewed and refined the workflow logic governing company transitions between different stages to ensure alignment with business requirements.
- Tested user interfaces and system functionalities after each modification to ensure stability and prevent regressions.
- Managed code changes using Git and pushed updates to GitHub through dedicated development branches.
- Developed a secure password reset feature that allows users to regain access to their accounts by entering their email address, receiving an OTP verification code, and creating a new password after successful verification. The implementation included dedicated interfaces for email submission, OTP verification, and password reset, providing a secure and user-friendly account recovery process while reducing account access issues.



- The database architecture of the **STEPIN** project was enhanced by migrating from **SQLite** to **PostgreSQL** using **Prisma ORM** as the data access layer. The work included reconfiguring the database connection, updating project settings to align with the new infrastructure, and ensuring data integrity and query compatibility throughout the migration process. This enhancement improved system performance, increased scalability, and prepared the project for deployment in production environments with greater reliability and efficiency.
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3. Skills and Knowledge Gained

- Improved and supported Localization and Internationalization (i18n) features to ensure a seamless multilingual user experience.
 - Analyzed software issues, identified bugs, and implemented effective solutions to improve system stability and performance.
 - Utilized Git and GitHub for version control, code management, and collaborative software development.
 - Designed and implemented automated workflows using the n8n automation platform.
 - Integrated Artificial Intelligence services with external applications to enhance system capabilities and automate processes.
 - Developed and integrated Telegram Bots with software systems to enable automated communication and workflow execution.
 - Conducted system testing and quality assurance activities to verify functionality and ensure reliable outputs before deployment.
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4. Challenges Faced

Multilingual Data Synchronization Challenges Within the System

As the system's functionality expanded and new features were introduced, several issues emerged related to multilingual data synchronization. Some newly added fields were not fully integrated with the localization files and database structure, resulting in incorrect values being displayed or certain information not appearing when users switched between the Arabic and English versions of the system.



5. Solutions and Improvements

The data flow between the database and the frontend was analysed, and all translation-related fields and localization keys were reviewed. Missing translation keys were added, localization files were updated, and the affected interfaces were properly linked to the translation resources. Functional testing was then conducted to verify that all data and text elements were displayed correctly in both languages, ensuring consistency and improving the overall user experience.

6. Conclusion

This training period significantly enhanced my technical skills through active participation in the development of a complete software system and by addressing real-world challenges related to databases, localization, and system integration. It also provided me with valuable hands-on experience in automation and artificial intelligence using the n8n platform and its integration with external services. These experiences strengthened my understanding of modern technologies and improved my ability to design and implement efficient, intelligent solutions in a professional work environment

End of Report

